

Ziyang Song

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Research Summary:

My research focuses on **AI for health**, with a focus on generative AI and statistical machine learning. My vision is to build **trustworthy LLMs** to address biomedical challenges.

1. Trustworthy LLM in healthcare: I design novel algorithms to enhance explainability, uncertainty analysis, and prediction safety for biomedical LLMs.
2. Generative AI for health time series: I focus on time-series representation learning for biosignals, clinical measurements, and longitudinal medical records.
3. Statistical machine learning for medical NLP: I design probabilistic models and statistical inference for interpretable medical representations using electronic health records.

Full-time Professional Experience:

- **Assistant Professor** 08/2025 – present
School of Electrical Engineering and Computer Science, Ohio University, Athens, Ohio, United States

Education:

- **Ph.D. Computer Science** 09/2020 – 05/2025
School of Computer Science, McGill University, Montreal, Canada
Affiliation: Mila - Quebec AI Institute
Advisor: [Yue Li](#)
- **M.Sc. Quality Systems Engineering** 09/2018 – 11/2019
Concordia Institute for Information Systems Engineering, Concordia University, Montreal, Canada
Advisor: [Nizar Bouguila](#)
- **B.Eng. Computer Science** 09/2014 – 06/2018
University of Shanghai for Science and Technology, Shanghai, China

Publications:

Conference Papers:

1. **Ziyang Song**, Qincheng Lu, Hao Xu, He Zhu, David Buckeridge, Yue Li. (2024). TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *ACM BCB, Rising Star Award*.
2. Zilong Wang, Ruohan Wang, **Ziyang Song**, David Buckeridge, Yue Li. (2024). MixEHR-Nest: Identifying subphenotypes within electronic health records through hierarchical guided-topic modeling. *ACM BCB*.
3. **Ziyang Song**, Qincheng Lu, He Zhu, David L. Buckeridge, Yue Li. (2024). Bidirectional generative pre-training for improving healthcare time-series representation *learning*. *MLHC (PMLR)*.
4. **Ziyang Song**, Ziyang Song, Yuanyi Hu, Aman Verma, David L. Buckeridge, Yue Li. (2022). Automatic phenotyping by a seed-guided topic model. *KDD, Health Day Best Paper award*.
5. **Ziyang Song**, Xavier Sumba Toral, Yixin Xu, Aihua Liu, Liming Guo, Guido Powell, Aman Verma, David Buckeridge, Ariane Marelli, Yue Li. (2021). Supervised multi-specialist topic model with applications on large-scale electronic health record data. *ACM BCB*.
6. **Ziyang Song**, Ornela Bregu, Samr Ali, Nizar Bouguila. (2019). Variational inference of finite asymmetric gaussian mixture models. *IEEE SSCI*.
7. **Ziyang Song**, Samr Ali, Nizar Bouguila. (2019). Bayesian learning of infinite asymmetric gaussian

mixture models for background subtraction. *ICIAR*.

Journal Papers:

1. **Ziyang Song**, Qincheng Lu, He Zhu, David Buckeridge, Yue Li. (2025). TrajGPT: Healthcare Time-Series Representation Learning for Trajectory Prediction. *IEEE Journal of Biomedical and Health Informatics (J-BHI)*.
2. **Ziyang Song**, Qincheng Lu, Hao Xu, Ziqi Yang, He Zhu, David Buckeridge, Yue Li. (2025). TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *Health Information Science and Systems*.
3. Yuesong Zou, Ahmad Pesaranghader, **Ziyang Song**, Aman Verma, David Buckeridge, Yue Li. (2022). Modeling electronic health record data using an end-to-end knowledge-graph-informed topic model. *Scientific Reports*.
4. **Ziyang Song**, Samr Ali, Nizar Bouguila. (2021). Bayesian inference for infinite asymmetric gaussian mixture with feature selection. *Soft Computing*.
5. **Ziyang Song**, Samr Ali, Nizar Bouguila. (2020). Background subtraction using infinite asymmetric Gaussian mixture models with simultaneous feature selection. *IET Image Processing*.
6. **Ziyang Song**, Samr Ali, Nizar Bouguila, Wentao Fan. (2020). Nonparametric hierarchical mixture models based on asymmetric gaussian distribution. *Digital Signal Processing*.

Preprints:

1. Jiahe Shen, He Zhu, **Ziyang Song**. (2025). SMIs: Semantic Medical IDs with Medical Ontology. Submitted to *AAAI SECURE-AI4H Workshop*.
2. **Ziyang Song**, Ziqi Yang, Shadi Zabad, Marc-André Legault, Yue Li. (2025). PheCode-guided multi-modal topic modeling of electronic health records improves disease incidence prediction and GWAS discovery from UK Biobank. Submitted to *Briefings in Bioinformatics*.

Presentations:

Oral Presentation:

1. **Ziyang Song**, et al. TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *ACM BCB*. Shenzhen, China. (2024, Nov 22-25).
2. **Ziyang Song**, et al. (2024, Sep 5-6). AI-driven approach for computational phenotyping with CARTaGENE cohort. *Scientific Meeting of the Canadian Translational Geroscience Network*. Montreal, Canada.
3. **Ziyang Song**, et al. (2024, Apr 8-13). Tutorial - phenotyping and PheWAS using MixEHR-seed. *Tokyo Symposium on Genomic Medicine, Therapeutics and Health*. Tokyo, Japan.
4. **Ziyang Song**, et al. (2023, May 15-18). MixEHR-SAGE: A seed-guided topic model to improve phenotyping for PheWAS analysis in UK Biobank data. *ISMB GLBIO*. Montreal, Canada.
5. **Ziyang Song**, et al. (2022, Aug 14-18). Automatic phenotyping by a seed-guided topic model. *KDD*. Washington DC, U.S.
6. **Ziyang Song**, et al. (2021, Aug 1-4). Supervised multi-specialist topic model with applications on large-scale electronic health record data. *ACM BCB*. Virtual.

Poster Presentation:

1. **Ziyang Song**, et al. (2024, Dec 15). TrajGPT: Healthcare Time-Series Representation Learning for Trajectory Prediction. *NeurIPS 2024 Workshop TSALM*. Vancouver, Canada
2. **Ziyang Song**, et al. (2024, Aug 16-17). Bidirectional generative pre-training for improving healthcare time-series representation learning. *MLHC*. Toronto, Canada
3. **Ziyang Song**, et al. (2022, Oct 21). MixEHR-Seed: automatic phenotyping by a seed-guided topic model. *50th Anniversary of SOCS at McGill University*. Montreal, Canada.
4. **Ziyang Song**, et al. (2019, Aug 27-29). Bayesian learning of infinite asymmetric gaussian mixture models for background subtraction. *ICIAR*. Waterloo, Canada

Honors and Awards:

Academic Honors:

- FRQNT Provincial Doctoral Research Scholarship 05/2023 – 09/2025
- McGill Research Stipend 09/2020 – 04/2024
- Grad Excellence Award in Computer Science 09/2020 – 04/2024
- Faculty of Science Grad Supplement Award 09/2022 – 08/2024
- SOCS Grad Stimulus Initiative Award 09/2020 – 08/2022
- Jackie Cheung Graduate Award 09/2020 – 08/2021
- Concordia Master Research Stipend 09/2018 – 10/2019

Awards:

- ACM BCB Rising Star Award 2024
- KDD Healthy Day Best Paper Award 2022

Internship Experience:

- **Research Intern** – Centre University Hospital (CHU) Sainte-Justine 06/2023 – 12/2024
 - Developed a clinical decision-support system using AI-driven models to track patient diagnoses, monitor disease progression, and analyze cancer registration.
 - Applied a probabilistic AI model (MixEHR-SAGE) to infer expert-guided phenotype distributions for 50K individuals in the CARTaGENE cohort. Presented in oral presentation.
- **International Research Intern** - Nanyang Technology University 01/2020 – 06/2020
 - Developed probabilistic models using stochastic optimization techniques to analyze retail data, optimizing product recommendations and promotions tailored to patient profiles.
 - Led to a significant improvement for item recommendations and patient-targeted promotions.
- **Data Analyst Intern** - Shanghai MetaLab 07/2017 – 12/2017
 - Developed a click-model for a recommender system and implemented machine learning algorithms to analyze patent data. Supported data analysis and visualization to aid in marketing decisions.
 - Developed a web crawler, database system, and back-end data interfaces to process around 10 million public patents. Maintained the database for efficient and reliable data analysis.
- **Data Analyst Intern** - Shanghai Qingyue Environment Protection Center (NGO) 01/2017 – 03/2017
 - Developed an environmental data platform that provides public access to weather data, aimed at supporting efforts to reduce environmental pollution in China.
 - Developed a web crawler, database system, and back-end data interfaces to efficiently process and manage publicly available weather datasets.

Teaching Experience

Ohio University

- **Instructor** Fall 2025
 - AI 5010/4010 Foundations of Deep Learning

McGill University

- **Guest Lecturer** Fall 2024
 - COMP 565 Machine Learning in Genomics and Healthcare
- **Guest Lecturer** Fall 2023
 - COMP 565 Machine Learning in Genomics and Healthcare
- **Teaching Assistant** Winter 2023
 - COMP 551 Applied Machine Learning
- **Teaching Assistant** Fall 2022
 - COMP 565 Applied Machine Learning

Supervision and Mentor Experience:

Ohio University (2025/07 - present)

PhD Students

- Long Nguyen Tan Starting from 01/2026

RA Students

- McMaster master student Hanwei Wu 05/2025 – present
 - Mentored in fine-tuning LLMs and Graph Machine Learning.
- UW–Madison master student Jiahe Shen 07/2025 – present
 - Mentored in applying semantic IDs in biomedical LLMs. Submitted to AAAI workshop 2026.
- Northwestern master student Zihao Li 07/2025 – present
 - Mentored in applying representation learning of medical concepts in hyperbolic spaces.
- Brown master student Yiran Ding 07/2025 – present
 - Mentored in applying constative learning and optimal transport in biomedical LLMs.
- OSU bachelor student Xiaohang Xie 07/2025 – present
 - Mentored in exploring confidence calibration in clinical reasoning and LLMs.
- OSU bachelor student Yang Hu 07/2025 – present
 - Mentored in exploring prediction confidence and rejection in LLMs.
- OSU bachelor student Zhongning Deng 07/2025 – present
 - Mentored in exploring uncertainty quantification and for LLM outputs.

McGill University (2020/09 - 2025/05)

- McGill master student Ruilin Wang 09/2024 – 2025/05
 - Mentored in applying LLMs in healthcare time-series data such as biosignals.
- McGill master student Bo Hong Wang 09/2024 – 2025/05
 - Mentored in applying LLMs in healthcare time-series data such as lab tests in ICU data.
- McGill undergraduate and master student Ziqi Yang 09/2022 – 2025/05
 - Mentored in applying MixEHR-Seed on UKB dataset, presented to ISMB GLBIO 2023.
 - Placement: master student, McGill University
- McGill undergraduate and master student Ruohan Wang 09/2022 – 09/2024
 - Mentored in designing MixEHR-Nest on Quebec’s PopHR dataset, published at ACM BCB 2024.
 - Placement: Ph.D. student, Brown University
- McGill undergraduate student Hao Xu 05/2023 – 09/2023
 - Mentored in applying TimelyGPT on biosignals, published at ACM BCB 2024.
 - Placement: Ph.D. student, University of Pennsylvania
- McGill undergraduate student Ziyu Zhao 05/2023 – 09/2023
 - Mentored in applying TimelyGPT on Quebec PopHR database.
 - Placement: master student, McGill University
- McGill undergraduate student Yuanyi Hu 05/2021 – 05/2022
 - Mentored in designing MixEHR-Seed, published at KDD 2022.
 - Placement: master student, Columbia University
- McGill undergraduate student Yixin Xu 09/2020 – 09/2021
 - Mentored in applying MixEHR-S, published at ACM BCB 2021.
 - Placement: master student, Duke University

Services:

Conference Reviewer:

- ICLR (2026, 2025)
- ACM SIGSPATIAL (2025)

- KDD (2025, 2024)
- ICML Workshop FMSD (2025)
- IJCAI (2025)
- NeurIPS Workshop TSALM (2024)
- ML4H (2024)

Journal Reviewer:

- Expert peer reviewer for BMJ Digital Health & AI (2025-2026)
- TMLR (2025)
- Applied Intelligence (2025)
- Big Data Mining and Analytics (2025, 2024)

Ohio University:

- Member of CS/AI Assessment & Accreditation (2025-2026)
- Member of PhD student committee (2025-present)

McGill University and Mila:

- Evaluation committee for Mila's Supervision Request Process (2024)

Referees:

- Dr. Yue Li. Associate Professor in School of Computer Science, McGill University.
send.Li.D38D6CE548@interfoliodossier.com
- Dr. David Buckeridge. Professor in School of Population and Global Health, McGill University.
send.Buckeridge.BC61C28AEC@interfoliodossier.com
- Dr. Archer Yi Yang. Associate Professor of Department of Mathematics and Statistics, McGill University.
send.Yang.BCEBBEFEC2@interfoliodossier.com